A07616000 - Richtpreisangebot

„Automatic production line for wire processing“

DRÄXLMAIER

30.09.2024

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Modulare Automation
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Gluth SYSTEM TECHNIK

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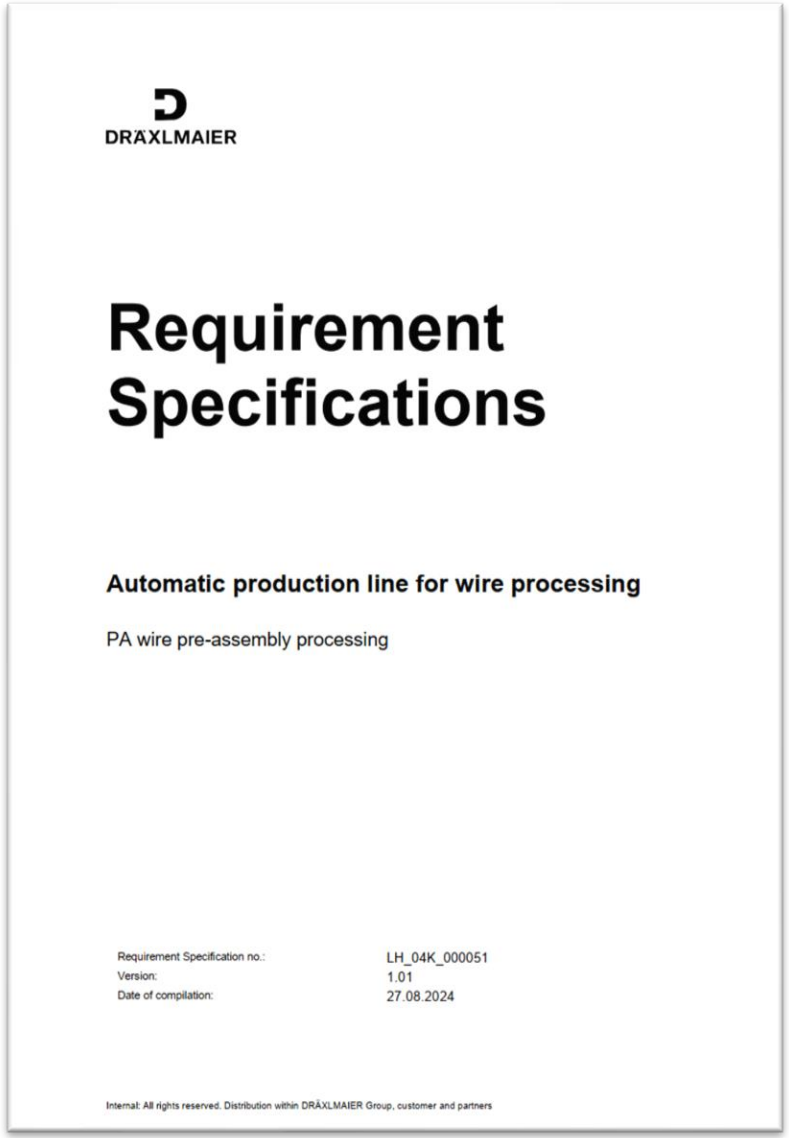
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Grundlagen

Lastenheft **„Automatic production line for wire processing“**
Version 1.01 vom 27.08.24

Taktzeit: < 25 sec.

Standort: Rumänien, Satu Mare



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Requirement Specifications

Automatic production line for wire processing

PA wire pre-assembly processing

Requirement Specification no.:	LH_04K_000051
Version:	1.01
Date of compilation:	27.08.2024

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Aufgabenbeschreibung

9.2 Product drawing

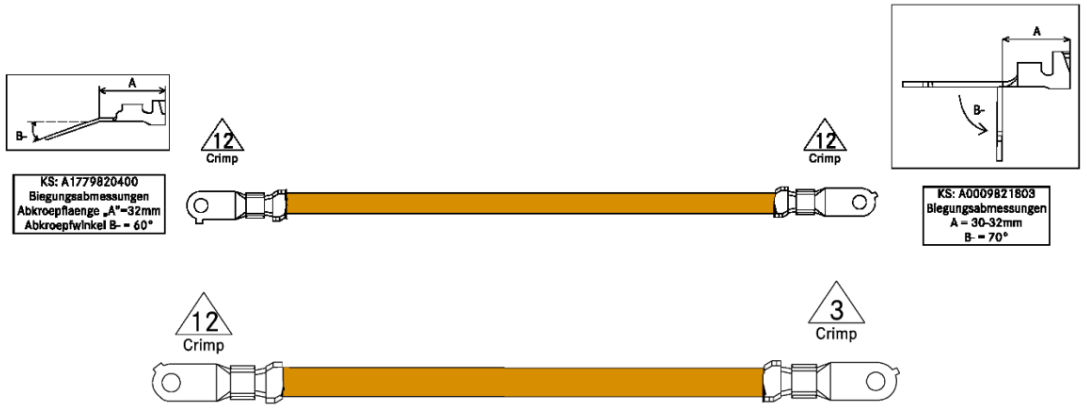
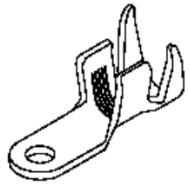
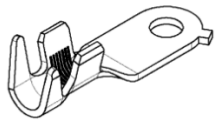


Figure 1: Assembled product

9.3 Assembly components

Designation	Amount	Material number
Cable (Low Voltage singlecore cable 16mm²/25mm² /35mm² FLRY-B;FLR2G;) 	1 pc	L0263CXX (16mm²) L3363066 (16mm²) L3366063 (25mm²) L3369048 (35mm²)
TB Cable lug, special shape (terminal)  	1 pc	A4813709 A4813711 A4813718 A4813719 A4813725 A4816710 A4816727 A4816743 A4817942 A4817952

Aufgabenbeschreibung

10 Process description

Process description

ST01 - Insertion station:
1. Manual scanning of the identification label placed on the cable into the DRX MES.
   - By scanning, the equipment receive the input data from DRX MES and load automatically the right processing program for the product variant
2. Manual insertion and positioning of the cable into the insertion unit (with possibility of buffer at least 20pcs).
   - In case that is processed the second end of cable must ensure the right orientation between the cable lugs on the two ends.
   - Insertion may only be possible in the correct position and the correct orientation (Poka Yoke)
3. Automatic fixing of cable by reaching the right position and orientation
4. Automatic checking of cable diameter correctness.
5. Automatic transfer of cable to the next station.

ST02 – Stripping station:
1. Automatic insert of cable end into the stripping unit.
2. Automatic stripping cut of insulation at the defined position.
3. Automatic removing of insulation part.
4. Automatic transfer of cable to the next station.



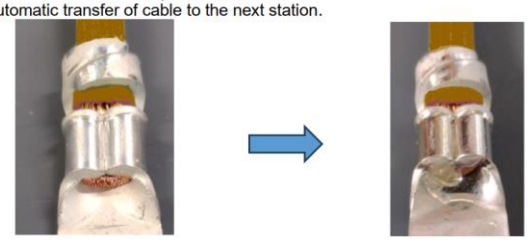
ST03 – Crimping station:
1. Automatic feeding of cable lug on the crimping position
2. Automatic insert of cable end into the crimping unit
3. Automatic crimping/pressing of cable lug on cable
4. Automatic transfer of cable to the next station.



ST04 – Bending station:
1. Automatic insert of cable end with crimped cable lug into the bending unit
   - If no bending is requested by the order, the product is transferred automatically to the next station.
2. Automatic bending of cable lug on the predefined position “A” to the defined angle “B” in the defined direction “+” or “-“.
3. Automatic transfer of cable to the next station.

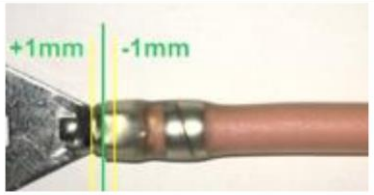


ST05 – Tinning station (soldering):
1. Automatic insert of cable end with crimped/bent cable lug into the tinning unit
2. Automatic tinning/soldering of the wires strands outcoming from crimped cable lug
3. Automatic cool down of product
4. Automatic transfer of cable to the next station.



ST06 – Quality inspection station:
1. Automatic insert of cable end with crimped/bent/tinned cable lug into the inspection unit
2. Automatic inspection of visual aspects of product
   - 360° vision system have to be implemented
3. Automatic transfer of cable to the next station.

☒ Option: ST07 – Shrinking station:
1. Automatic positioning of shrink tube on the cable lug
   - If no shrinking is requested by the order, the product is transferred automatically to the next station.
2. Automatic insert of cable end with crimped/bent/tinned cable lug into the shrinking unit
3. Automatic contraction of the shrink tube
4. Automatic cool down of product
5. Automatic transfer of cable to the next station.



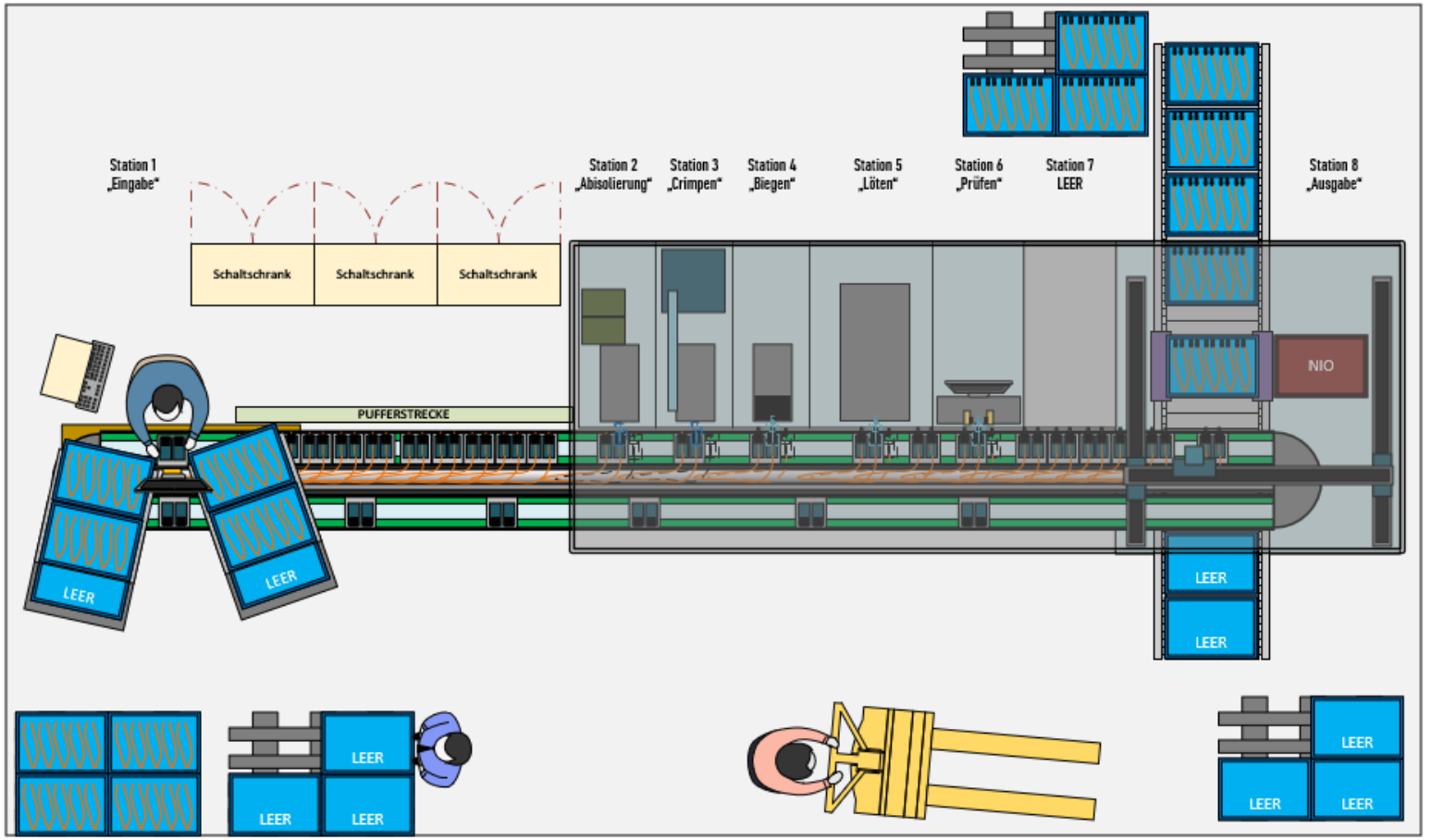
ST08 – Output station:
1. Automatic release of OK product and place on the defined position (collecting recipient)
2. Automatic separation of NOK product in the defined position (NOK box)

Variante 1

Variante 1:
„Vollautomatisiert“
Vorgaben laut Lastenheft

Eingabe Station 1: Manuell
Station 2 – 8: Automatisch

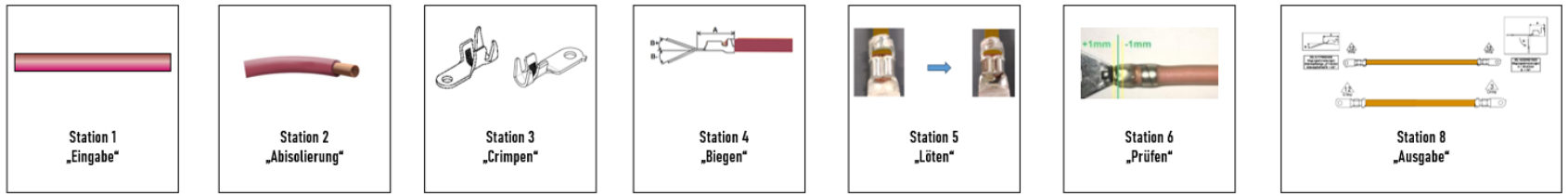


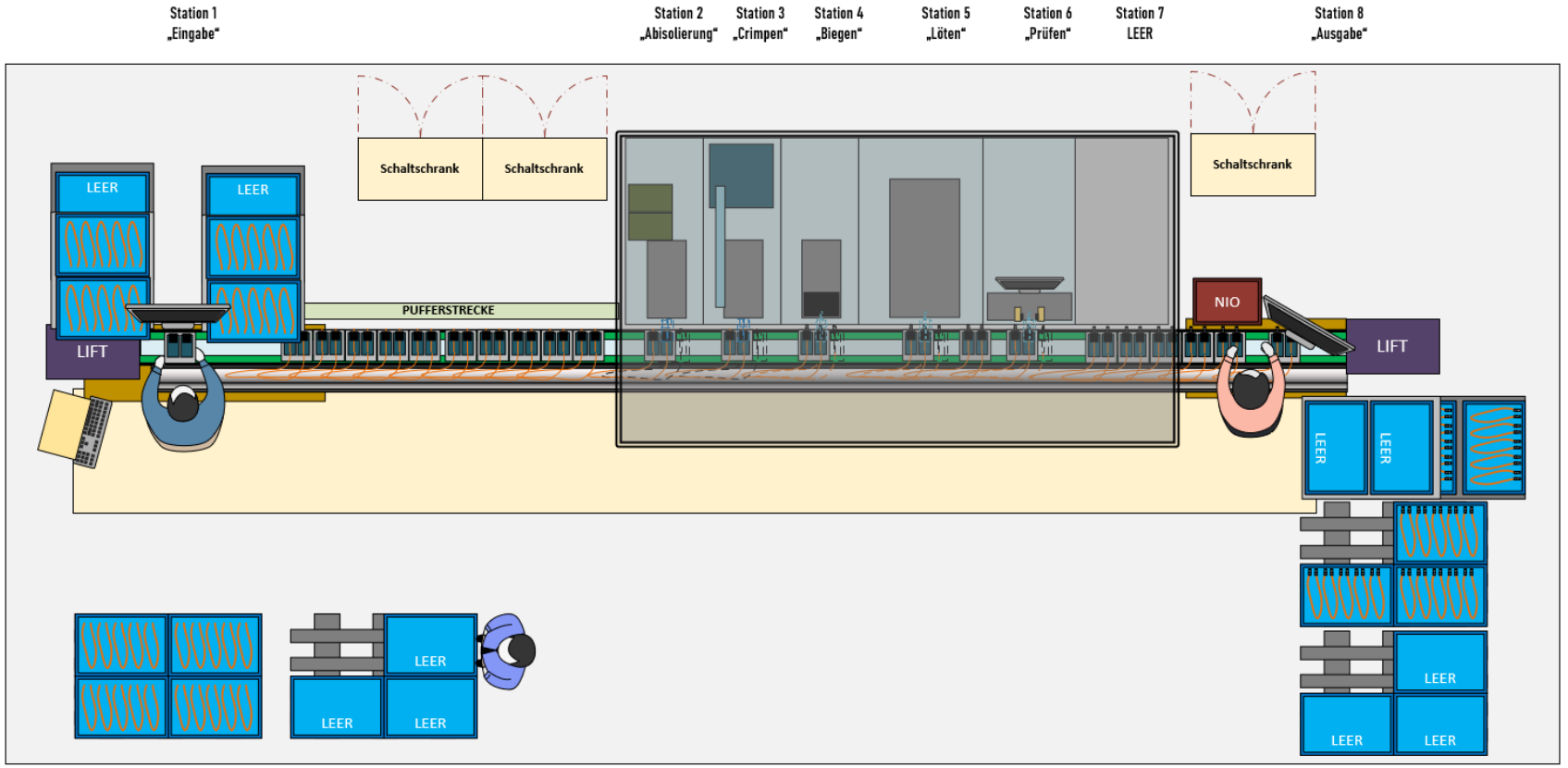


Variante 2

Variante 2:
„Teilautomatisiert“

Eingabe Station 1: Manuell
Station 2 – 6: Automatisch
Ausgabe Station 8: Manuell

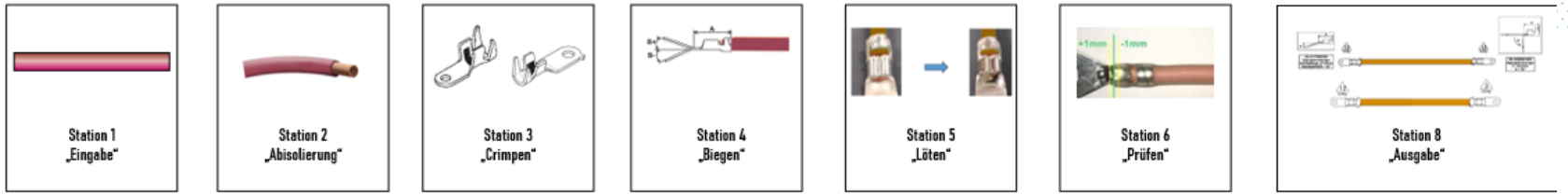


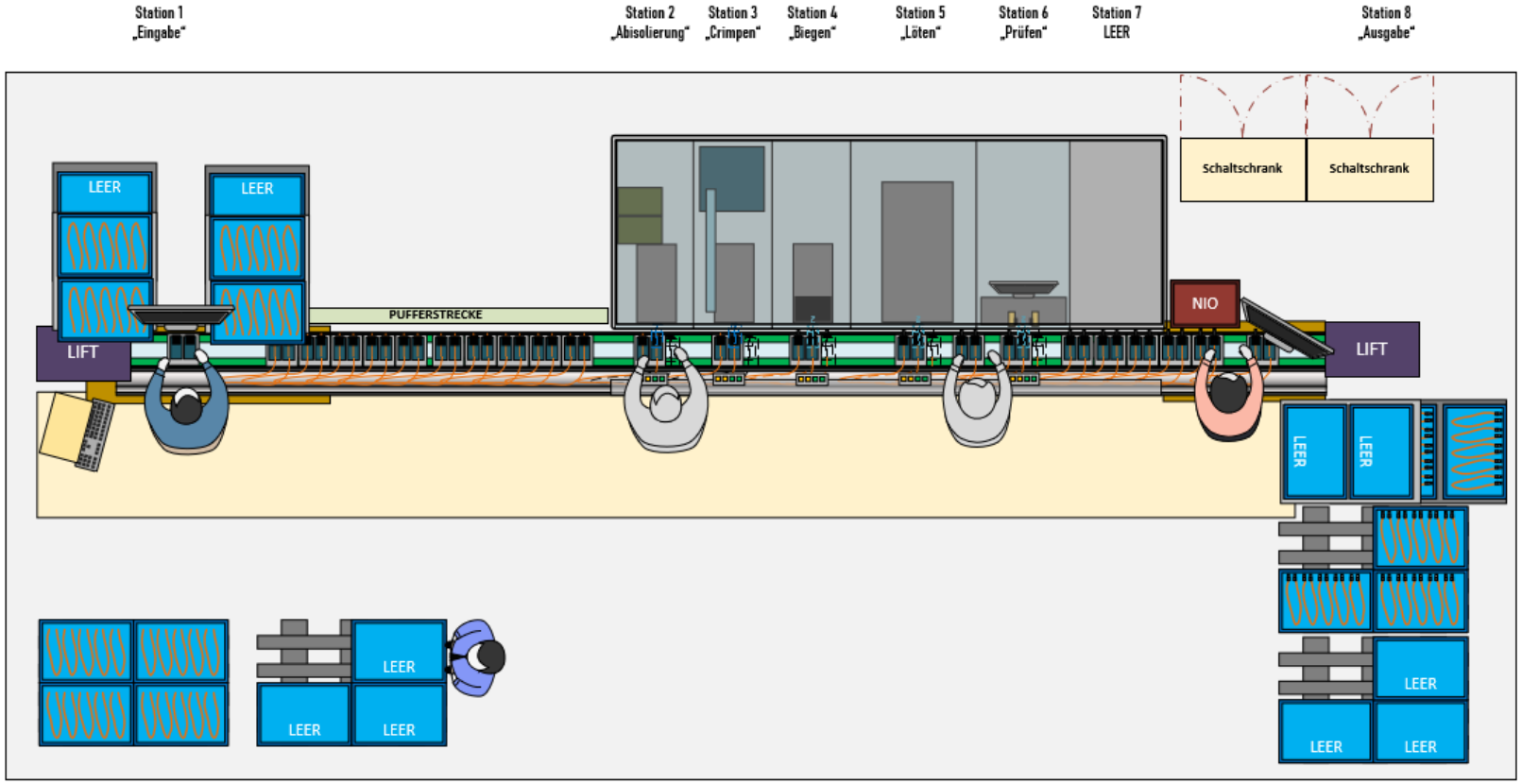


Variante 3

Variante 3:
„Teilautomatisiert mit Handzuführungen“

Eingabe Station 1: Manuell
Station 2 – 6: Zuführung Manuell mit Automatischen Prozessen
Ausgabe Station 8: Manuell







Danke
für die Aufmerksamkeit

Gluth SYSTEM TECHNIK

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